

DE Lab1
Hayden Fuller

Q1: HJS P2.1

a) VFB and VT0

NMOS VFB = -1.05

PMOS VFB = 0.93

NMOS Vt0 = -0.0185

PMOS Vt0 = -0.138

NMOS Vt0 = VFB + |2phiF| - QB0/Cox = -0.0185

PMOS Vt0 = VFB - |2phiF| - QB0/Cox = -0.138

NMOS VFB = phiGC - Qox/Cox = -1.05

PMOS VFB = phiGC - Qox/Cox = 0.93

NMOS phiGC = phiG - phiC = -0.99V

PMOS phiGC = phiG - phiC = 0.99V

NMOS phiG = half bandgap = -.55V

NMOS phiC = phiF = 0.44V

PMOS phiG = half bandgap other way = .55V

PMOS phiC = -phiF? = -0.44V

Qox/Cox = 0.0611 C/F = 0.061V

Qox = qNf C/cm^2 = 9.6x10^-8 C/cm^2

q = 1.6x10^-19 C

Nf = 6x10^11 cm^-2 Process Parameters

Cox = epsilonox/tox = 1.57 uF/cm^2

epsilonox = 4e0 = 3.5x10^-13 F/cm

tox = 22Angstroms = 220nm = 2.2x10^-7 cm Process Parameters

100nm >> 34.50 nF/cm^2

2.2nm >> 34.5/2.2*100 = 1.57 uF/cm^2

phiF = kT/q ln(ni/p) = 0.44V

kT/q = 26 mV

p = NA = 3x10^17 cm^-3

NMOS QB0/Cox = -0.189

PMOS QB0/Cox = 0.189

NMOS QB0 = -sqrt(2q epsilonSi NA |2 phiF|) = -2.96x10^-7 C/cm^2

PMOS QB0 = sqrt(2q epsilonSi NA |2 phiF|) = 2.96x10^-7 C/cm^2

q = 1.6x10^-19 C

epsilonSi 1.04x10^-10 F/m = 1.04x10^-12 F/cm

NA = 3x10^17 cm^-3

phiF = 0.44V

Cox = 1.57 uF/cm^2

b) PMOS but donors?

ϕ_{iG} flips sign

$\phi_{iG} = \text{half bandgap other way} = -0.55\text{V}$ (down 1.1V from before)

$\phi_{iGC} = \phi_{iG} - \phi_{iC} = -0.55 - (-0.44) = -0.11\text{V}$ (down 1.1V from before)

$V_{FB} = \phi_{iGC} - Q_{ox}/C_{ox} = -0.11 - 0.06 = -0.17$ (down 1.1V from before)

$V_{T0} = V_{FB} - |2\phi_{iF}| - Q_{B0}/C_{ox} = -1.238$ (down 1.1V from before)

i) It drops by double ϕ_{iG} since its sign is flipped and it's a direct positive 1 to 1 correlation

ii) $V_{T0} = -1.238\text{V}$

c) $\phi_{iG} = 0$?

ϕ_{iG} becomes 0

NMOS $V_{T0} = V_{FB} + |2\phi_{iF}| - Q_{B0}/C_{ox} = 0.5315$ (up .55)

NMOS $V_{FB} = \phi_{iGC} - Q_{ox}/C_{ox} = -0.50$

NMOS $\phi_{iGC} = \phi_{iG} - \phi_{iC} = -0.44\text{V}$

NMOS $\phi_{iG} = 0$ (up .55)

NMOS $\phi_{iC} = \phi_{iF} = 0.44\text{V}$

NMOS $V_{T0} = 0.5315$

PMOS $V_{T0} = V_{FB} - |2\phi_{iF}| - Q_{B0}/C_{ox} = -0.688$ (down .55)

PMOS $V_{FB} = \phi_{iGC} - Q_{ox}/C_{ox} = 0.38$

PMOS $\phi_{iGC} = \phi_{iG} - \phi_{iC} = 0.44\text{V}$

PMOS $\phi_{iG} = 0$ (down .55)

PMOS $\phi_{iC} = -\phi_{iF} = -0.44\text{V}$

PMOS $V_{T0} = -0.688$

d) $V_{T0}=0.3\text{V}$, what implant concentrations?

$Q_{imp} = q N_{imp} x_{imp}$

$\Delta V_T = -Q_{imp}/C_{ox}$

NMOS $N_{imp} = -\Delta V_T C_{ox} / (q x_{imp}) = -1.25 \times 10^{18} \text{ cm}^{-3}$

PMOS $N_{imp} = -\Delta V_T C_{ox} / (q x_{imp}) = 6.3585 \times 10^{17} \text{ cm}^{-3}$

$n^+ \text{PMOS } N_{imp} = -\Delta V_T C_{ox} / (q x_{imp}) = -3.68165 \times 10^{18} \text{ cm}^{-3}$

$C_{ox} = 1.57 \text{ uF/cm}^2$

$q = 1.6 \times 10^{-19} \text{ C}$

$x_{imp} = 25 \text{ nm} = 2.5 \times 10^{-6} \text{ cm}$

NMOS $\Delta V_T = 0.3 - (-0.0185) = 0.3185$

PMOS $\Delta V_T = -0.3 - (-0.138) = -0.162$

$n^+ \text{PMOS } \Delta V_T = -0.3 - (-1.238) = 0.938$

NMOS $N_{imp} = -1.25 \times 10^{18} \text{ cm}^{-3}$

PMOS $N_{imp} = 6.3585 \times 10^{17} \text{ cm}^{-3}$

e) why n^+ NMOS p^+ PMOS?

because this gate choice minimizes the threshold voltage without requiring large, unrealistic channel implants.

Q2: HJS P2.3

i) long or short?

$W=L=100\text{nm}$

1.2V

$\epsilon_{\text{silicon}} = 1.2\text{V}/100\text{nm} = 1.2 \times 10^5 \text{ V/cm} = 120\text{kV/cm}$

short because it reaches carrier velocity saturation:

$\epsilon_{\text{silicon}} > E_c = 6 \times 10^4$

ii) NMOS

$$y_0 = 24 \cdot 10^{-6} \cdot e^{\left(-\frac{3}{0.028}\right)} \cdot \left(1 - e^{-\frac{x}{0.028}}\right)$$

$$y_4 = \left\{ x < x_4: \frac{2(167 \cdot 1.57 \cdot 10^{-6})}{1 + \frac{x}{6}} \left(0.4 - .3 - \frac{x}{2}\right) x, x > x_4: 10^{-5} \cdot 10^7 \cdot 1.57 \cdot 10^{-6} \cdot \frac{(0.4 - 0.3)^2}{(0.4 - 0.3) + .6} \right\}$$

$$x_4 = \frac{(0.4 - .3) \cdot .6}{(0.4 - .3) + .6}$$

$$= 0.0857142857143$$

$$y_8 = \left\{ x < x_8: \frac{2(167 \cdot 1.57 \cdot 10^{-6})}{1 + \frac{x}{6}} \left(0.8 - .3 - \frac{x}{2}\right) x, x > x_8: 10^{-5} \cdot 10^7 \cdot 1.57 \cdot 10^{-6} \cdot \frac{(0.8 - 0.3)^2}{(0.8 - 0.3) + .6} \right\}$$

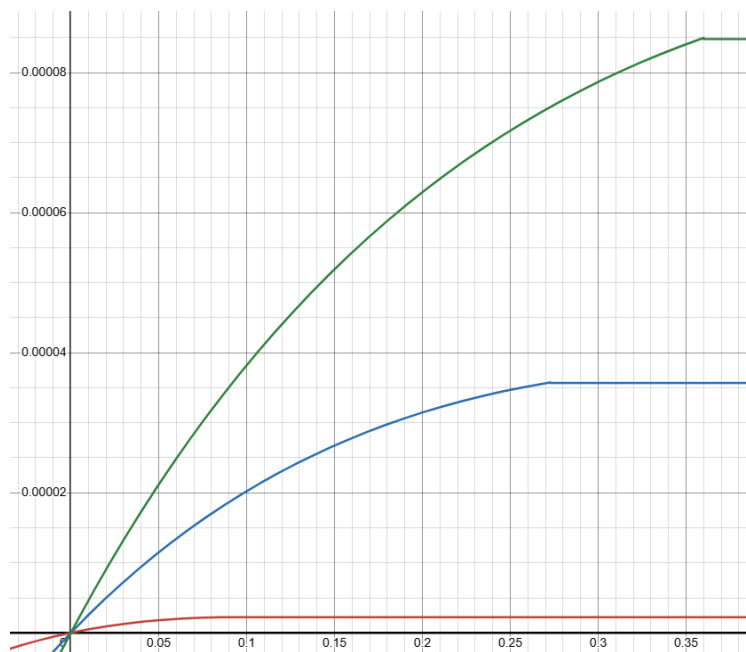
$$x_8 = \frac{(0.8 - .3) \cdot .6}{(0.8 - .3) + .6}$$

$$= 0.272727272727$$

$$y_{12} = \left\{ x < x_{12}: \frac{2(167 \cdot 1.57 \cdot 10^{-6})}{1 + \frac{x}{6}} \left(1.2 - .3 - \frac{x}{2}\right) x, x > x_{12}: 10^{-5} \cdot 10^7 \cdot 1.57 \cdot 10^{-6} \cdot \frac{(1.2 - 0.3)^2}{(1.2 - 0.3) + .6} \right\}$$

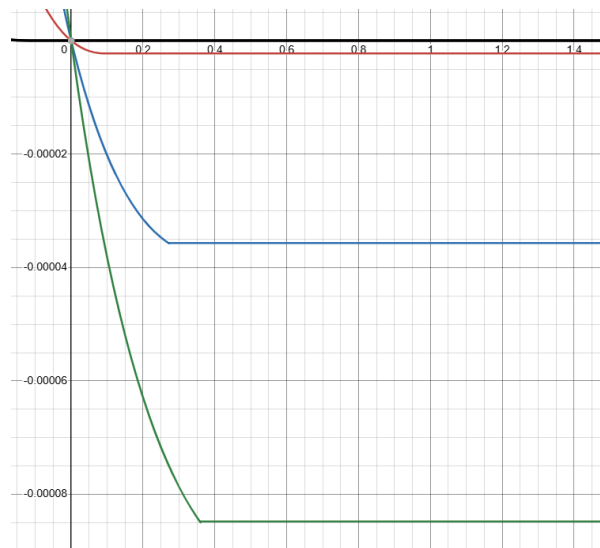
$$x_{12} = \frac{(1.2 - .3) \cdot .6}{(1.2 - .3) + .6}$$

$$= 0.36$$



They get the right idea across, but I'm 90% sure I have some numerical errors here

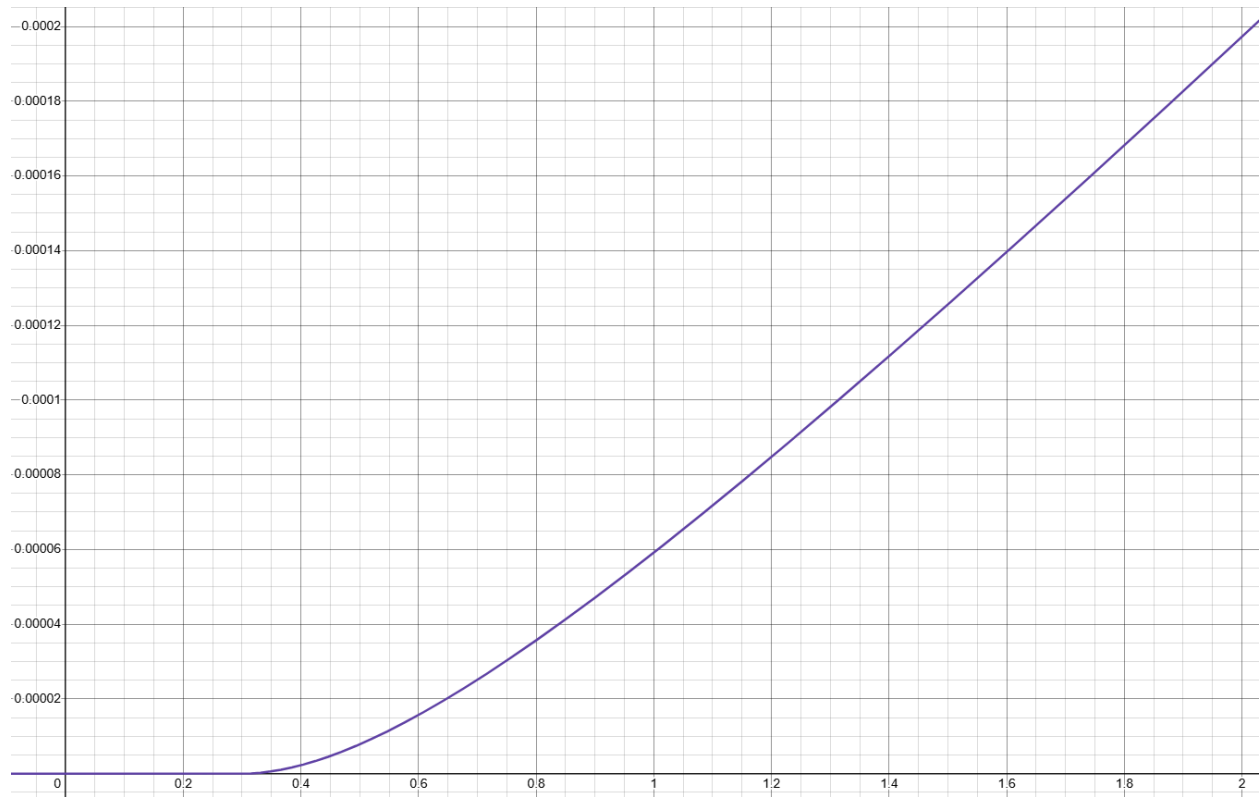
PMOS



I'm sure this has the same numerical errors, I'll continue to work on this, but midnight is approaching

iii)

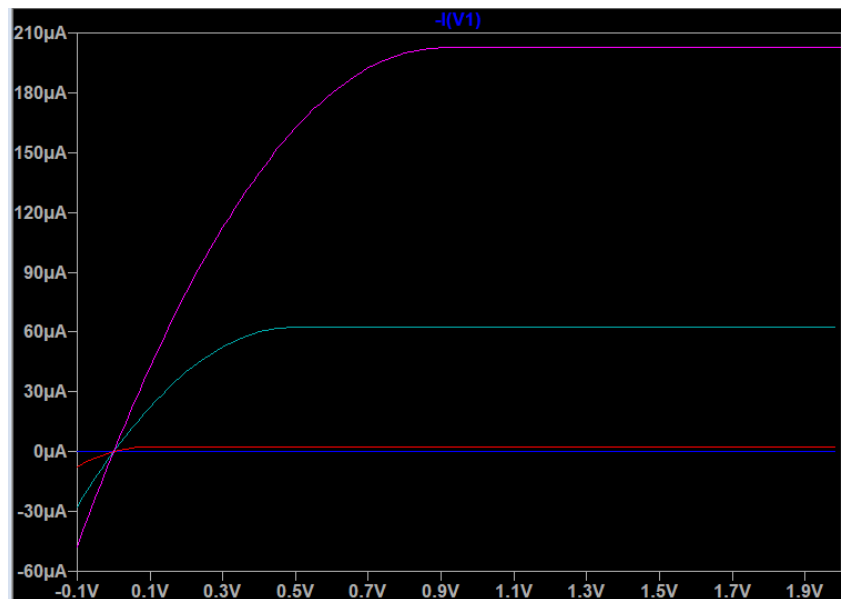
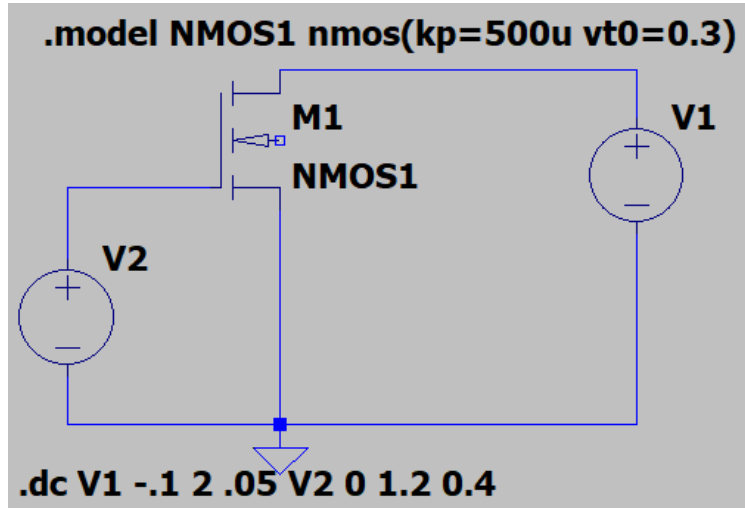
$$y = \begin{cases} x < .3: 0, x > .3: 10^{-5} \cdot 10^7 \cdot 1.57 \cdot 10^{-6} \cdot \frac{(x - 0.3)^2}{(x - 0.3) + .6} \end{cases}$$



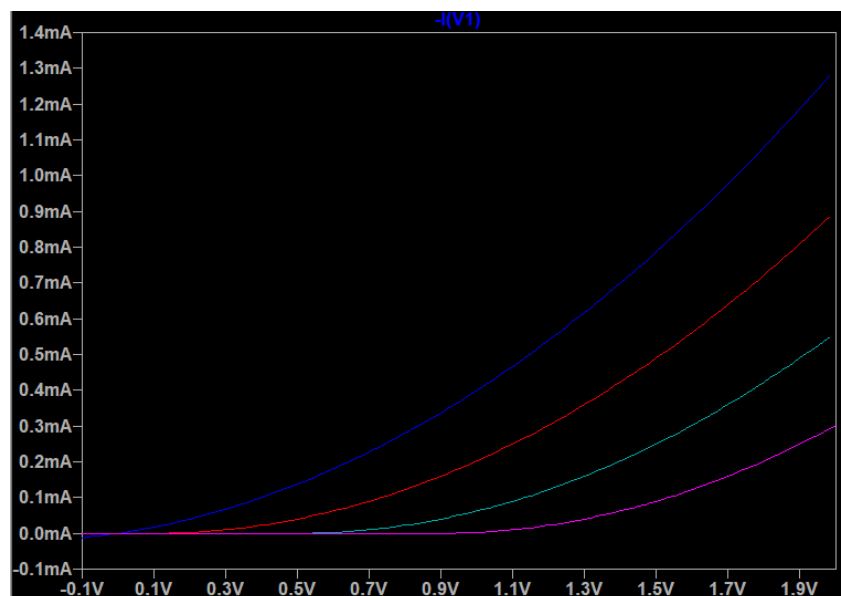
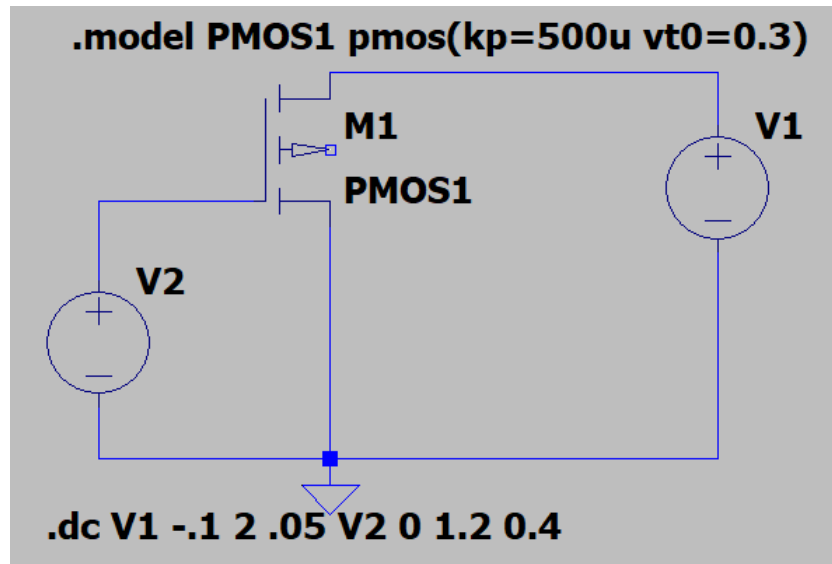
I believe this is more accurate than part ii, it seems much closer to linear than quadratic

iv) I think you meant parts ii and iii? can't really compare to spice for i?
 note, I didn't see this comparison until it was too late, and I think we used $V_{t0}=0.3V$ in class? but the simulation page says to use $0.4V$? I'm sticking with $0.3V$ in the simulations to (hope to) match my calculations

iv - ii)

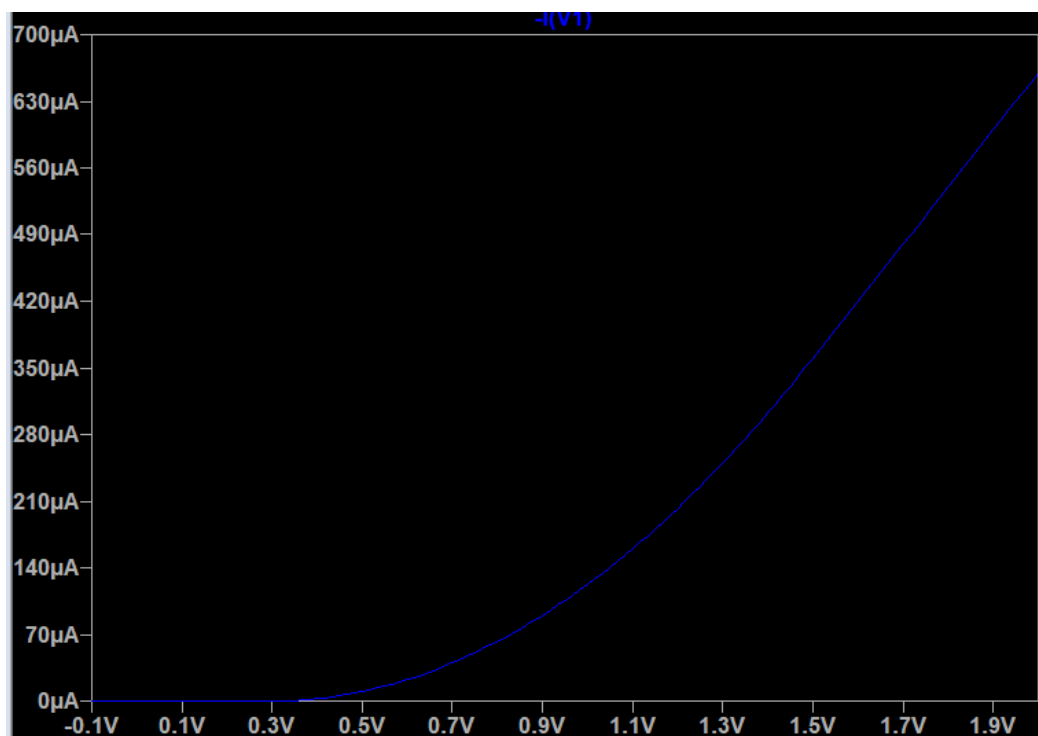
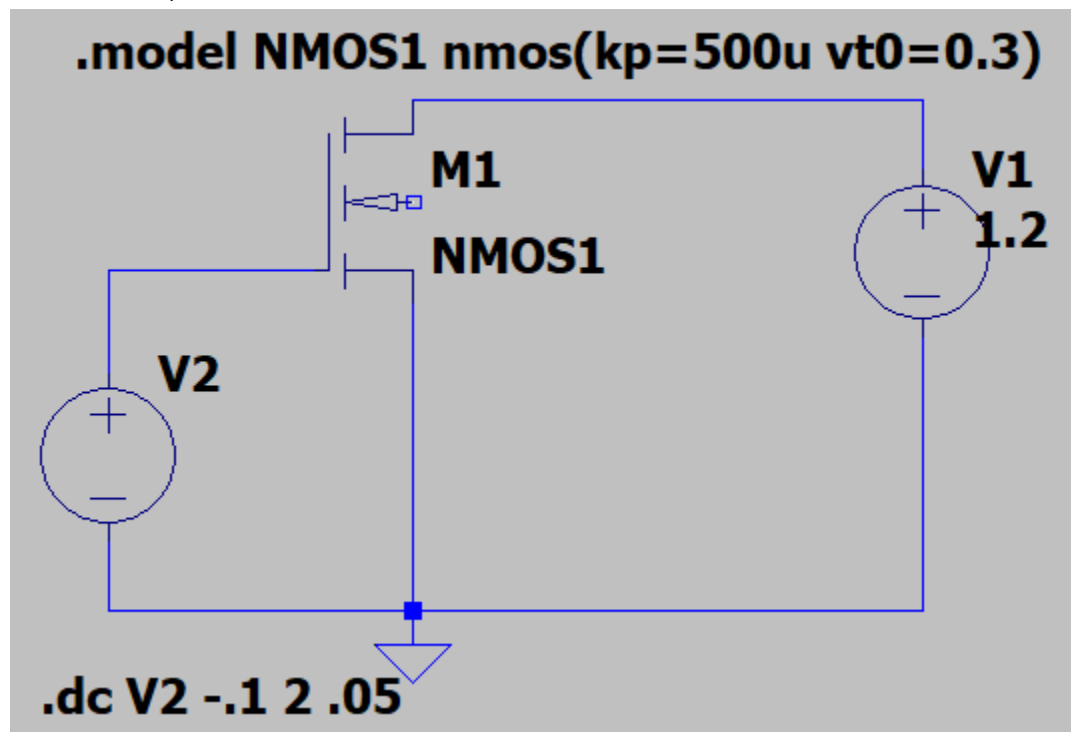


These generally match up with my hand calculations in terms of shape and trends, I'm still messing with calculations as I write this, but my V_{Dsat} and corresponding saturation currents are much lower, the currents by a factor of 2.3885



The PMOS simulation is, I don't think what is expected. I probably set something up wrong here? not sure what, I'll continue to troubleshoot it, but I've been working on this since office hours this morning and it's nearly midnight, I'll try again tomorrow

iv - iii)



My scale is off again, lower current by a factor of about 3, but otherwise relatively accurate, I'll keep messing with it.

Q3: HJS P2.5)

a)

V_{Dsat} at $V_{GS}=1.2V$ is $.36V$

$$((V_{GS}-V_T) E_c L) / ((V_{GS}-V_T) + E_c L) = ((1.2-.3) .6) / ((1.2-.3) + .6)$$

$V_{DS}=1.2$ is in saturation region

V_x rises, V_{GS} drops until $V_{T0} = .4$

$$V_{GS} = 1.2 - V_x = .4$$

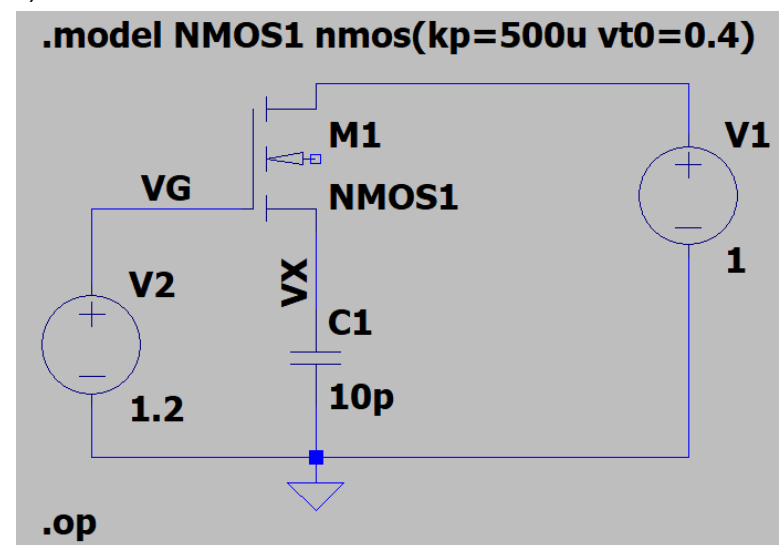
$$V_x = 0.8V$$

$V_{DS}=0.6$ is still in saturation region

V_x still maxes at $0.8V$, but only $0.6V$ is supplied

$$V_x = 0.6V$$

b)

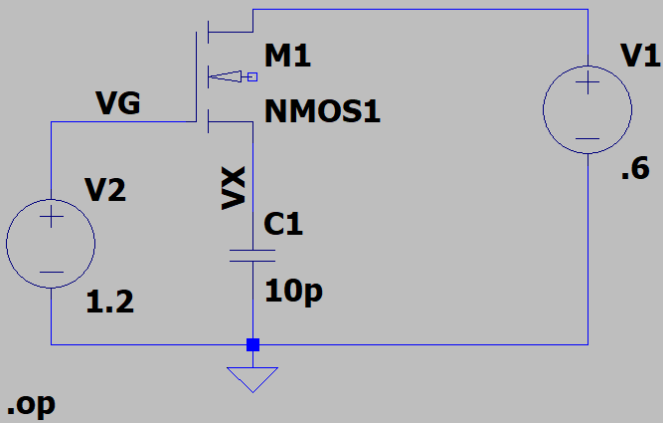


C:\Users\Hayden\Documents\LTspice\Draft1.net

--- Operating Point ---

V(n001):	1	voltage
V(nc_01):	0.606498	voltage
V(vg):	1.2	voltage
V(vx):	0.7995	voltage
I(C1):	7.995e-24	device_current
I(V1):	-6.06498e-13	device_current
I(V2):	0	device_current
Ib(M1):	-6.06498e-13	device_current
Id(M1):	6.31055e-11	device_current
Ig(M1):	0	device_current
Is(M1):	-6.2499e-11	device_current

.model NMOS1 nmos(kp=500u vt0=0.4)



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--- Operating Point ---

V(n001):	0.6	voltage
V(nc_01):	0.406663	voltage
V(vg):	1.2	voltage
V(vx):	0.6	voltage
I(C1):	6e-24	device_current
I(V1):	-4.06663e-13	device_current
I(V2):	0	device_current
Ib(M1):	-4.06663e-13	device_current
Id(M1):	4.09833e-13	device_current
Ig(M1):	0	device_current
Is(M1):	-3.17022e-15	device_current

they match!